Sarawak Campus

Assignment Cover Sheet

(for individual and group assignments)

This cover sheet is to be attached to all assignments, both hard copy and electronic format

ASSIGNMENT DETAILS

Unit Code	FSE10023	Unit Title	Linear algebra
Tutorial/Lab Group	Group 2	Lecturer/Tutor Name	Chai Ching Yi
Assignment Title	FSE10023 Linear Algebra Assignment [Semester 2 2026]		
Due date	09/30/2026	Date Received	09/24/2026

DECLARATION

For both individual and group assignments, in the case of assignment submission on behalf of another student, it is assumed that permission has been given. The University takes no responsibility for any loss, damage, theft, or alteration of the assignment.

To be completed if this is an individual assignment

I declare that this assignment is my individual work. I have not worked collaboratively, nor have I copied from any other student's work or from any other source/s, except where due acknowledgment is made explicitly in the text, nor has any part been written for me by another person.

Student Details	Student ID Number	Student Name	Student Signature
Student 1			

To be completed if this is a group assignment

We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.

Student Details	Student ID Number(s)	Student Name(s)	Student Signature (s)
Student 1	Wong Ik Chan	105811359	Ik Chan
Student 2	Ethan Liew Kin You	105809493	Ethan Liew
Student 3			
Student 4			
Student 5			

MARKER'S COMMENTS

Total Mark		Marker's Signature		Date	

EXTENSION CERTIFICATE

This assignment has been given an extension by

Unit Convenor			
Extended due date		Date Received	

Declaration and Statement of Authorship

1. I/we have not impersonated, or allowed myself/ourselves to be impersonated by any person for the purposes of this assessment.
2. This assessment is my/our original work and no part of it has been copied from any other source except where due acknowledgement is made.
3. No part of this assessment has been written for me/us by any other person except where such collaboration has been authorised by the lecturer/teacher concerned.
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Task 1

i. Number of cars, Number of parking remaining

ii. Velocity, tension

Task 2

i. Distance of Pole B, $|\vec{OB}| = \sqrt{(-2)^2 + 3^2}$

$|\vec{OB}| = \sqrt{13}$

Reference angle, $\beta = \tan^{-1}\left(\frac{3}{2}\right)$

$\beta = 56.31°$

$\theta = 180° - 56.31°$

$\theta = 123.69°$

ii. $\vec{OB} - \vec{OA} = (-2\hat{i} + 3\hat{j}) - (3\hat{i} - 2\hat{j})$

$= (-2-3)\hat{i} + (3-(-2))\hat{j}$

$= -5\hat{i} + 5\hat{j}$

Distance, $|\vec{OB} - \vec{OA}| = \sqrt{(-5)^2 + 5^2}$

$|\vec{OB} - \vec{OA}| = 5\sqrt{2}$

Reference angle, $\beta = \tan^{-1}\left(\frac{5}{5}\right)$

$\beta = 45°$

$\theta = 180° - 45°$

$\theta = 135°$

Task 3

$\overrightarrow{OP} = 4\hat{i} + 3\hat{j}$

$\overrightarrow{OQ} = -3\hat{i} + 2\hat{j}$

$\overrightarrow{OP} \cdot \overrightarrow{OQ} = (4 \times -3) + (3 \times 2)$

$= -6$

$\therefore -6$

Task 4

$$-3\vec{OC} + \frac{1}{3}\vec{OD} = -3(6\hat{i} + 4\hat{j}) + \frac{1}{3}(-12\hat{i} + 9\hat{j})$$

$$= (-18\hat{i} - 12\hat{j}) + (-4\hat{i} + 3\hat{j})$$

$$= (-18 + (-4))\hat{i} + (-12 + 3)\hat{j}$$

$$= (-22\hat{i} - 9\hat{j})N$$

Task 5

$\overrightarrow{OE} = 2\hat{i} + \hat{j}$

$\overrightarrow{OF} = 3\hat{i} - 2\hat{j} + \hat{k}$

$\overrightarrow{OG} = 3\hat{j} - \hat{k}$

$\overrightarrow{OE} \times \overrightarrow{OF}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 0 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= \hat{i} - 2\hat{j} + (-4-3)\hat{k}$$

$$= \hat{i} - 2\hat{j} - 7\hat{k}$$

$(\hat{i} - 2\hat{j} - 7\hat{k}) \times (3\hat{j} - \hat{k})$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -7 \\ 0 & 3 & -4 \end{vmatrix} = (8 + 21)\hat{i} - (-4\hat{j}) + 3\hat{k}$$

$\therefore 29\hat{i} + 4\hat{j} + 3\hat{k}$

Task 6

i. $|\overrightarrow{OH}| = \sqrt{4^2+5^2+7^2}$

$|\overrightarrow{OH}| = 3\sqrt{10}$

$\cos\alpha = \frac{4}{3\sqrt{10}}$

$\cos\beta = \frac{5}{3\sqrt{10}}$

$\cos\gamma = \frac{7}{3\sqrt{10}}$

ii. $\alpha = \cos^{-1}\left(\frac{4}{3\sqrt{10}}\right)$

$\alpha = 65.1°$

$\beta = \cos^{-1}\left(\frac{5}{3\sqrt{10}}\right)$

$\beta = 58.2°$

$\gamma = \cos^{-1}\left(\frac{7}{3\sqrt{10}}\right)$

$\gamma = 42.5°$

Task 7

$d(6,2)$

$F(40,30)$

$W = Fd$

$= (6,2) \cdot (40,30)$

$= (6 \times 40) + (2 \times 30)$

$= 240 + 60$

$= 300\ J$

Task 8

$$\vec{OP} \times \vec{OQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & 0 \\ 1 & 5 & 0 \end{vmatrix}$$

$$\vec{OP} \times \vec{OQ} = (0-0)\hat{i} - (0-0)\hat{j} + (20-1)\hat{k}$$

$$\vec{OP} \times \vec{OQ} = 19\hat{k}$$

$$|\vec{OP} \times \vec{OQ}| = \sqrt{19^2}$$

$$|\vec{OP} \times \vec{OQ}| = 19$$

Area of triangle $= \frac{1}{2}(19)$

$= 9.5\ m^2$